PHILIPPINE SOCIETY OF ANIMAL NUTRITIONISTS

39th ANNUAL CONVENTION

Fueling Progress: Shaping Animal Nutrition for a Competitive and Sustainable Future

6 OCTOBER 2026

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PHILIPPINE SOCIETY OF ANIMAL NUTRITIONISTS
39th Annual Convention | October 6, 2026

Next Generation of Animal Nutrition: Enhancing Feed Value Through Canadian Peas

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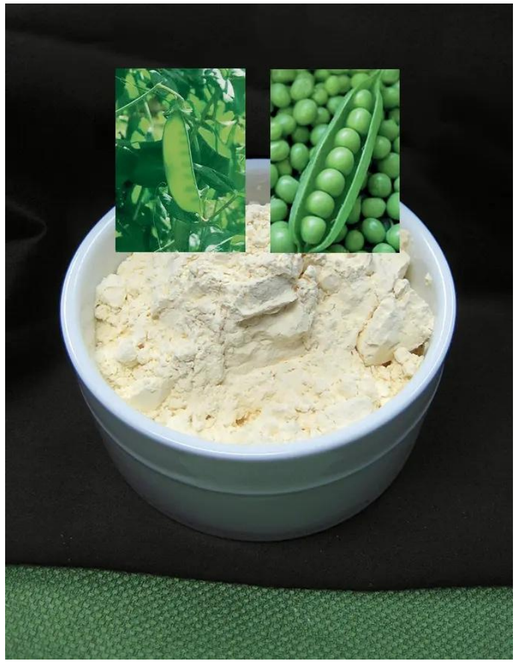


FIELD PEAS

- Peas (*Pisum sativum*): one of the four most important legumes, after soybeans, groundnuts, and beans.
- Peas are commonly grown in temperate areas of the globe and across a wide range of environments.
- Yellow peas are used most often in swine and layer feeds.

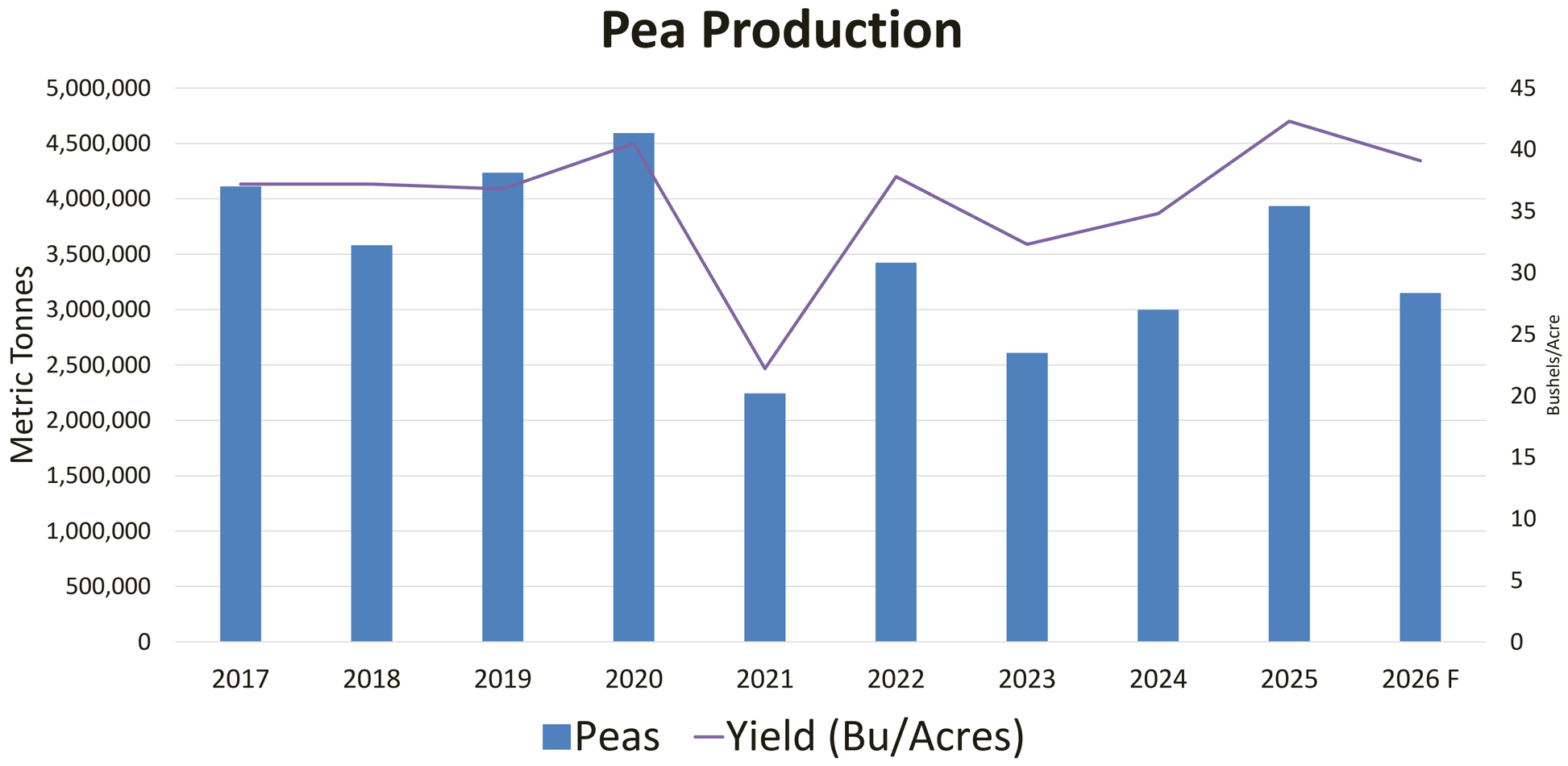
PEAS *(Pisum sativum):* A FOOD AND FEED INGREDIENT

- ***CANADA:*** **major producer of green and yellow peas; largest world exporter.**
- ***EUROPE, NORTH AMERICA:*** **mainly used in animal feed.**
- ***ASIA, SOUTH AMERICA:*** **mainly used as food; also used in animal feed.**
- ***PEA PROTEIN:*** **highly digestible; no trypsin inhibitors; low levels of other anti-nutritional factors.**





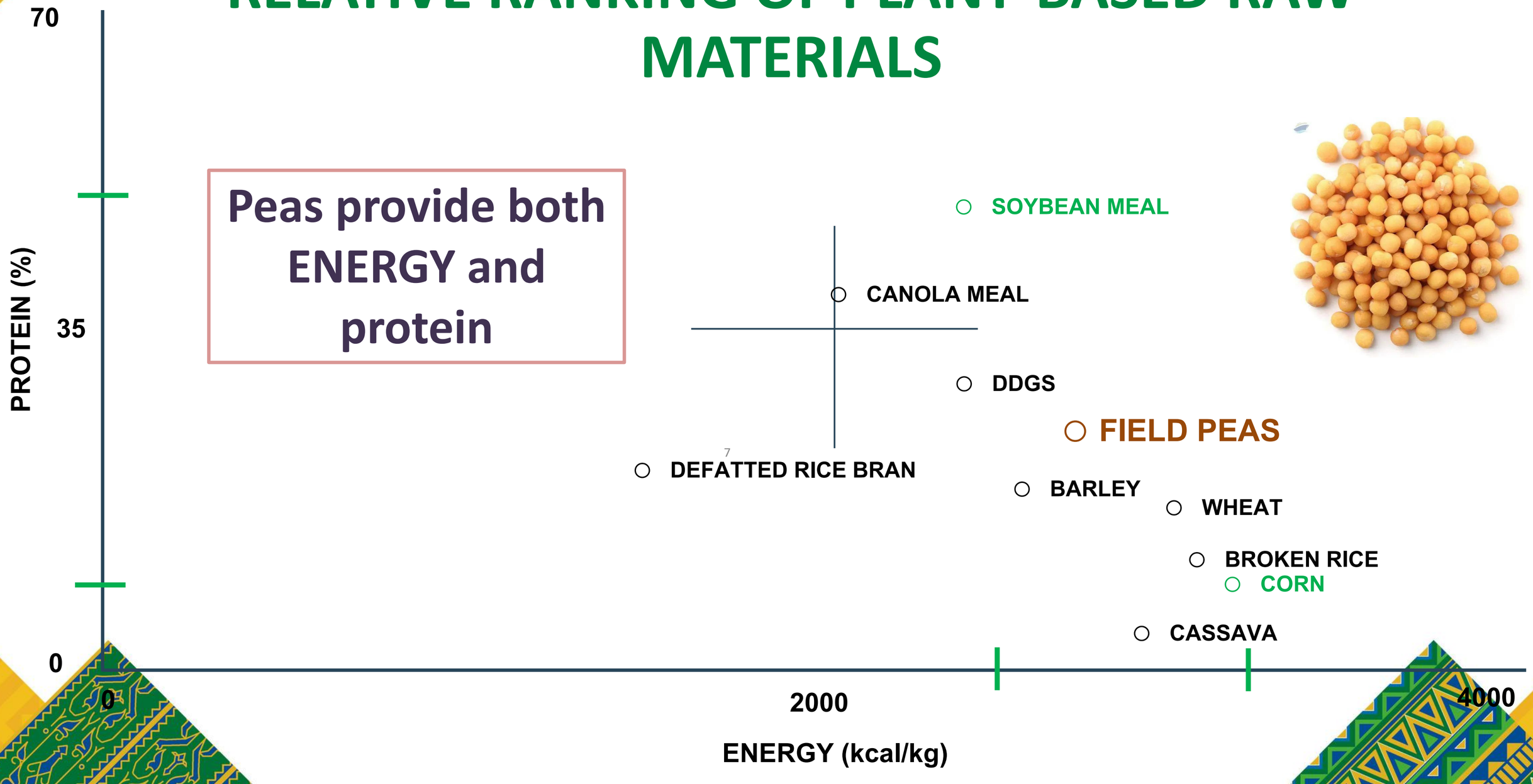
CANADA PEA PRODUCTION



GLOBAL PEA PRODUCTION

	2021	2022	2023	2024	2025
Australia	261,140	313,809	**228,498**	**230,000**	**250,000**
Canada	2,244,000	3,423,000	**2,609,000**	**2,997,354**	**3,900,000**
United States	416,000	702,000	**818,000**	**761,000**	**950,000**
France	552,000	400,000	**487,000**	**456,000**	**500,000**
Russia	3,168,000	3,616,000	**4,700,000**	**4,000,000**	**5,400,000**
Ukraine	578,000	375,000	**384,000**	**360,000**	**550,000**
India Sub-Cont.	874,000	890,000	**858,000**	**868,000**	**1,000,000**
Other	3,227,000	3,701,000	**3,137,000**	**3,216,000**	**3,150,000**
TOTAL	11,320,140	13,420,809	**13,221,498**	**12,888,354**	**15,800,000**

RELATIVE RANKING OF PLANT-BASED RAW MATERIALS



Peas provide both ENERGY and protein

PROXIMATE ANALYSIS *(as-fed)*

Nutrient	Peas	Corn	Wheat	Soybean Meal	Canola Meal
Crude Protein (%)	22.2	8.5	13.3	48.1	37.8
Crude Fat (%)	1.2	3.8	2.5	1.0	3.8
Crude Fiber (%)	6.2	2.2	3.0	3.9	12.0
Starch (%)	43.5	64.1	55.5	0.4	1.8
Ash (%)	2.9	1.2	1.9	6.4	8.6
DE Swine (Kcal/kg)	3504	3600	3550	3700	3150
AMEn (kcal/kg)	2805	3350	2900	2440	2000

Peas are a moderate source of protein and energy, making them a suitable ingredient in most animal and aquatic feeding programs.

USE OF PEAS in SWINE FEEDING PROGRAMS



GROWTH PERFORMANCE OF VIETNAMESE PIGS WHEN PEAS REPLACE SOYBEAN MEAL

	T1 0% pea	T2 25% pea	T3 50% pea	T4 75% pea	T5 100% pea
Initial weight (kg)	60.73	60.67	60.71	60.70	60.73
Final weight (kg)	106.43	106.39	106.30	105.36	106.28
Average Daily Gain (g)	**815.92**	**816.37**	**814.06**	**797.62**	**813.39**
Daily Feed Intake (kg)	**2.19**	**2.19**	**2.19**	**2.20**	**2.20**

The effect of replacing soybean meal by field pea (*Pisum sativum*) seed meal in diets on growth performance, carcass traits and meat quality of finishing pigs in the tropical climate conditions of Viet Nam

Nguyen Vu Thuy Hong Loan, La Van Kinh, William Riley[1], La Thi Thanh Huyen[2] and Le Duc Ngoan[3]

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VIETNAM SWINE TRIAL

Demonstrated that Yellow Peas can totally replace SBM and be included in finisher pig diets at up to 31% with no adverse effect on growth performance or meat characteristics while potentially improving feed cost.

SAMPLE FORMULATIONS FOR SWINE

Ingredients	Lactating Sow	Grower	Finisher
Corn	455	430	550
DDGS	0	30	25
SBM (47.5%)	175	30	35
Cottonseed meal	0	40	0
Sunflower meal	0	40	0
Vitamin-mineral premix	50	40	40
Wheat bran	50	30	20
Soybean oil	30	20	30
Fish meal	20	40	0
Yellow peas	220	300	300
Total	1000	1,000	1000
Corn/DDGS/SBM*	65/0/35	63/0/37	60/15/25
ME, kcals/kg	3,350	3,250	3,400
Crude Protein, %	18.5	16.0	13.5
*% Replacement by yellow peas			

USE OF PEAS in POULTRY FEEDING PROGRAMS

- Broilers
- Laying Hens
- Ducks



YELLOW PEA INCLUSION IN LAYING HEN DIETS: IMPACT ON PERFORMANCE AND EGG QUALITY

Dr. Chanwit Kaewtapee, Department of Animal Science, Kasetsart University, Bangkok, Thailand





- **Test Product: Yellow peas**
- **Duration: 12 weeks (84 days)**
- **Species: Layers**
- **Breed: Hisex brown**
- **Sex: Female**
- **Age of Animals: 49 - 61 weeks**
- **Housing: Battery Cage (40 × 40 × 30 cm)**





PRODUCTION PERFORMANCE

PARAMETER	CONTROL	8% YELLOW PEAS	16% YELLOW PEAS	24% YELLOW PEAS	P-VALUE
Body weight (g)	1,861±33	1,863±15	1,877±64	1,871±31	0.892
Hen-day egg production (g/d)	92.68±2.64	93.92±1.79	91.20±4.48	90.98±2.54	0.323
Feed intake (g/d)	108.67±1.99	108.91±2.11	108.88±4.27	110.12±1.33	0.774
Egg mass (g/d)	56.74±1.32	58.34±2.43	56.52±3.44	57.33±1.93	0.573
Feed conversion ratio (FCR)	1.92±0.05	1.87±0.06	1.94±0.11	1.93±0.08	0.549

Production performance was unaffected at all inclusion levels, with all parameters showing no significant differences between any treatment group and the control group across the full 12-week trial (P>0.05).

Values are means SEM (n=6 replicates of 9 birds). *Significant differences (p<0.05).
Superscripts ^{a,b} denote differences between means within rows (Tukey HSD).

EGG QUALITY AND BLOOD MARKERS

PARAMETER	CONTROL	8% YELLOW PEAS	16% YELLOW PEAS	24% YELLOW PEAS	P-VALUE
Egg weight (g)	61.27±1.05	62.06±1.78	61.92±1.48	62.95±0.55	0.200
Albumen weight (g)	40.50±0.83	41.23±1.22	40.94±1.16	41.69±0.37	0.211
Yolk weight (g)	14.59^{b}±0.32	14.57^{b}±0.38	$14.80^{a,b}$±0.25	15.06^{a}±0.15	0.026*
Yolk colour (Roche scale)	7.44^{a}±0.07	7.43^{a}±0.14	7.41^{a}±0.16	7.19^{b}±0.08	0.005*
Haugh unit	90.94±0.95	90.94±1.41	91.54±0.78	91.87±0.93	0.345
Shell thickness (mm)	0.47±0.01	0.47±0.01	0.47±0.01	0.46±0.01	0.217
WBC (10^3/µL)*	10.60^{a}±1.80	10.23^{a}±1.66	5.52^{b}±1.70	5.40^{b}±0.77	<0.001*
Heterophil (10^3/µL)	7.00^{a}±1.87	6.47^{a}±1.30	3.63^{b}±1.32	3.47^{b}±0.70	<0.001*
Heterophyll/Lymphocyte	2.32±0.98	2.02±0.65	2.30±0.55	2.08±0.62	0.849

Values are means±SEM (n=6 replicates of 9 birds). *Significant differences. Superscripts ^{a,b} denote differences between means within rows (Tukey HSD). *WBC: white blood cell count

USE OF PEAS IN AQUACULTURE FEEDING PROGRAMS





VIETNAM SHRIMP FEEDING TRIAL*

GROWTH PERFORMANCE OF SHRIMP AFTER 56-DAYS FEEDING

Parameters	Control	5%	10%	15%
Feed Intake (g)	12.00^{a}	11.99^{a}	12.02^{a}	12.02^{a}
Weight Gain (g)	10.72^{b}	10.26^{ab}	10.02^{ab}	9.48^{a}
FCR	1.12^{a}	1.17^{ab}	1.21^{ab}	1.27^{b}
Specific Growth Rate (% day)	3.37^{a}	3.30^{a}	3.27^{a}	3.20^{a}
Survival Rate (%)	90.0^{a}	91.1^{a}	88.9^{a}	90.0^{a}

FCR: Feed conversion ratio

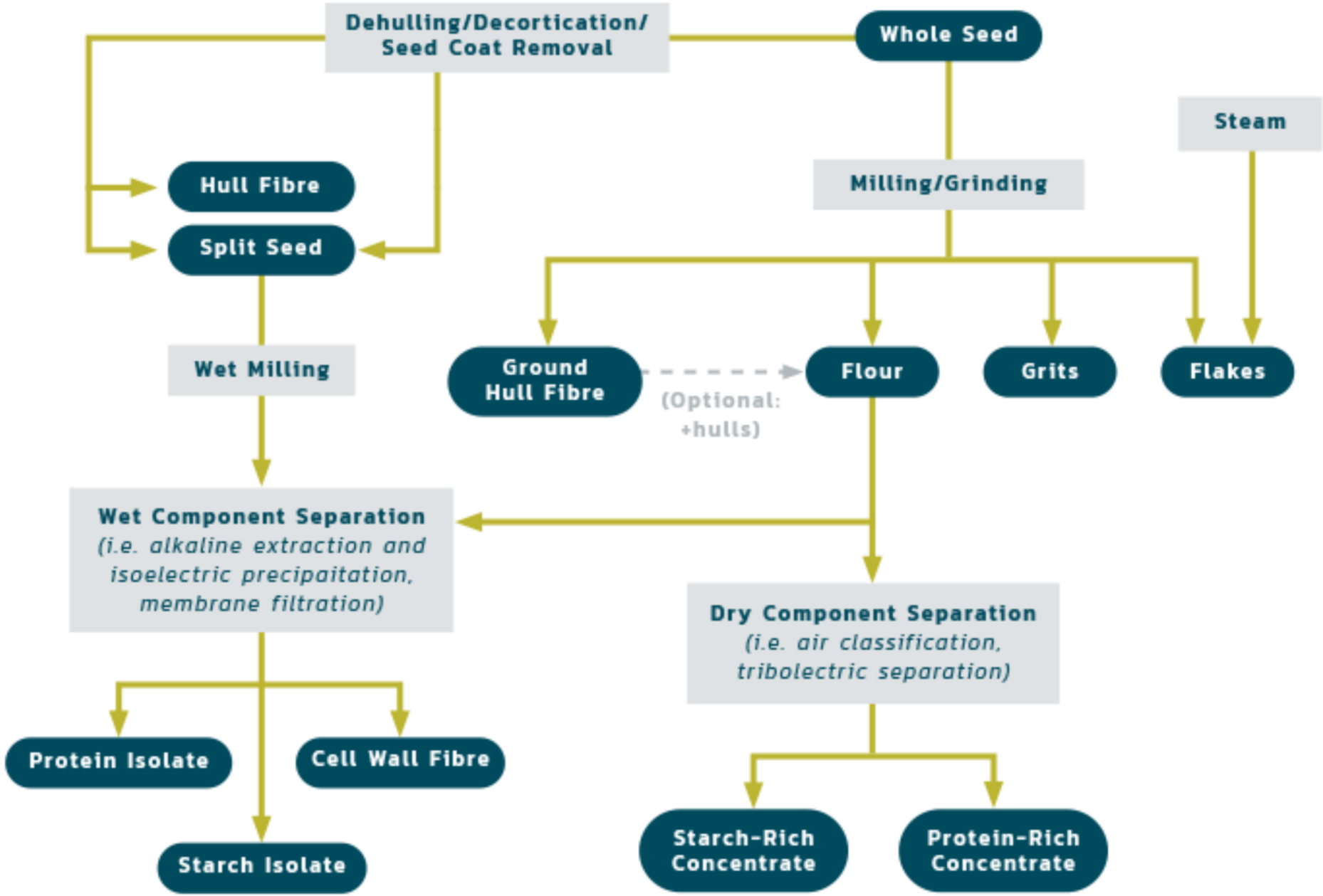
*Nguyen et al. (2022). Research Center for Aquafeed Nutrition and Fishery Post-Harvest Technology (APOTEC)- Research Institute for Aquaculture No. 2, Ho Chi Minh, Vietnam

YELLOW PEA FRACTIONS

- Available as concentrates of starch, protein, or fibre (hulls); mainly used in FOOD applications, but available for animal use as well.
- Produced via air classification (milling) and aqueous extraction (filtration).
- For feed, fractions offers nutritive value, pellet binding characteristics, and functional benefits (prebiotic).
- Limited use in poultry with more value in pig starter, AQUACULTURE, and PETFOOD applications.



VALUE ADDED PROCESSING OF PEAS



PEA STARCH CONCENTRATE: AN ALTERNATIVE TO WHEAT FLOUR IN THE DIET OF WHITE LEG SHRIMP

Rex Ferdinand Traifalgar, Ph.D., College of Fisheries and Ocean Sciences, University of the Philippines Visayas, Iloilo, Philippines

1. **Evaluate the optimum inclusion level of *Pea Starch Concentrate (PSC)* as a substitute for *Wheat Flour* on growth performance, feed conversion efficiency, and immune response of White Leg Shrimp *(Penaeus vannamei).***
2. **Document the influence of dietary inclusion levels of *PSC* on the carcass composition of juvenile White Leg Shrimp .**
3. **Determine dry matter and carbohydrate apparent digestibility coefficients for White Leg Shrimp as an ingredient in shrimp feed.**

EXPERIMENTAL PROTOCOL

DIET	CONTROL	TRT 2	TRT 3	TRT 4	TRT 5
Wheat flour replaced, %	0	25	50	75	100
Pea starch concentrate in diet, %	0	5	10	15	20
Wheat flour in diet, %	20	15	10	5	0

- Five isonitrogenous, isocaloric diets (CP 40%, DE 14 MJ/kg), meeting NRC requirements for *P. vannamei.*
- Basal diet contained fish meal (CP 70%), squid meal, wheat gluten, SBM (48%), yeast, fish oil, cholesterol, lecithin (70%), vitamin-mineral premix, and BHT.
- Diets fed at 3% bodyweight, 4-times daily for 8-weeks to 300 juvenile White Leg shrimp in 15-tanks, 3-tanks per treatment (20-shrimp per tank).

INGREDIENT NUTRIENT COMPOSITION

AS FED, %	WHEAT FLOUR	PEA STARCH CONCENTRATE
Dry Matter	88.0	90.0
Crude Protein	11.7	12.0
Carbohydrate	70-75	60 (as starch)
Crude Lipid	1.2	3.0
Ash	0.4	4.0

PEA STARCH CONCENTRATE APPARENT DIGESTIBILITY*

NUTRIENT	DIGESTIBILITY, %
Dry Matter	90.25±2.78
Protein	95.01±0.70
Lipid	90.89±2.31
Carbohydrate	95.95±1.00

*Apparent digestibility was assessed against a reference diet using 1% chromium oxide as an inert marker.

GROWTH PERFORMANCE

	0% * 0% **	25% 5%	50% 10%	75% 15%	100% 20%
Survival, %	88.33±2.89	86.67±5.77	86.67±2.89	88.33±2.89	90.00±0.00
Weight Gain, %	2087.05±63	1934.92±262	1942.88±148	2041.60±329	2083.28±158
Specific Growth Rate (SGR), % day^{-1}	3.41±0.13	3.28±0.25	3.27±0.17	3.30±0.26	3.37±0.15
Feed Conversion Ratio (FCR)	1.77±0.08	1.83±0.06	1.89±0.09	1.85±0.16	1.89±0.08
Feed Intake (FI), g/day	13.66±0.74	13.03±1.70	13.41±1.97	13.43±2.71	14.26±1.47

*Percentages denote wheat flour replacement level (top) and **actual percentage of pea starch concentrate in treatment diets (bottom). Values are mean±SEM (n=3 replicate tanks). Survival, weight gain, SGR, FCR, and FI were statistically comparable across all treatments (P>0.05).

INTESTINAL MORPHOLOGY

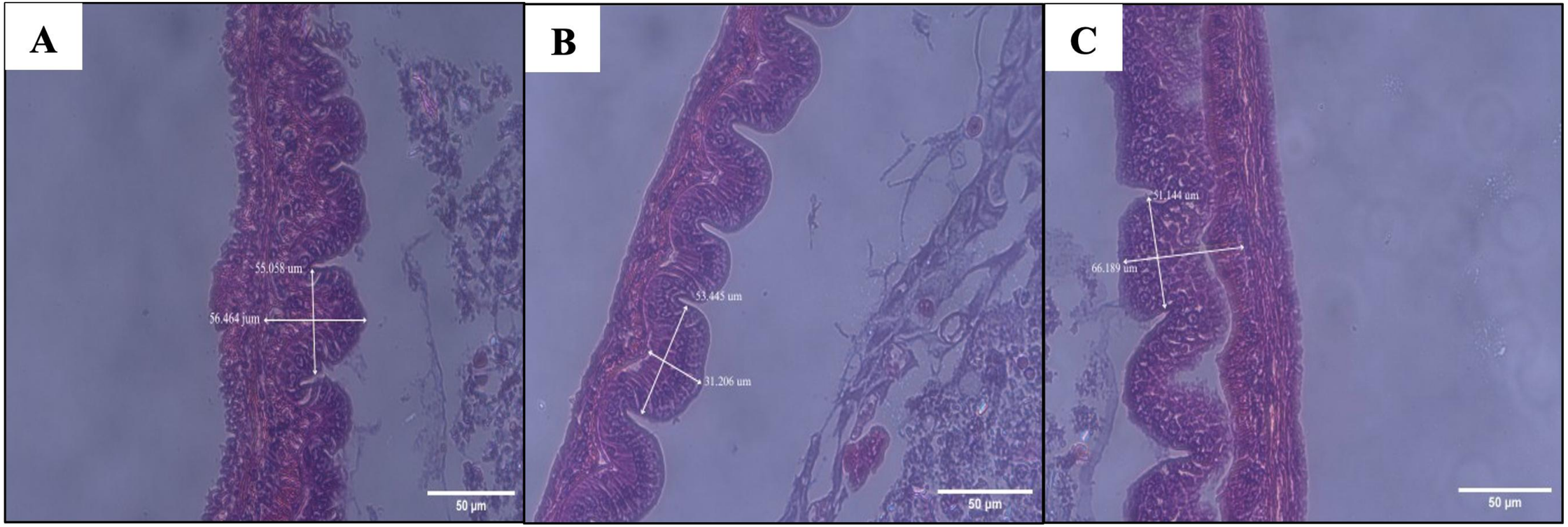
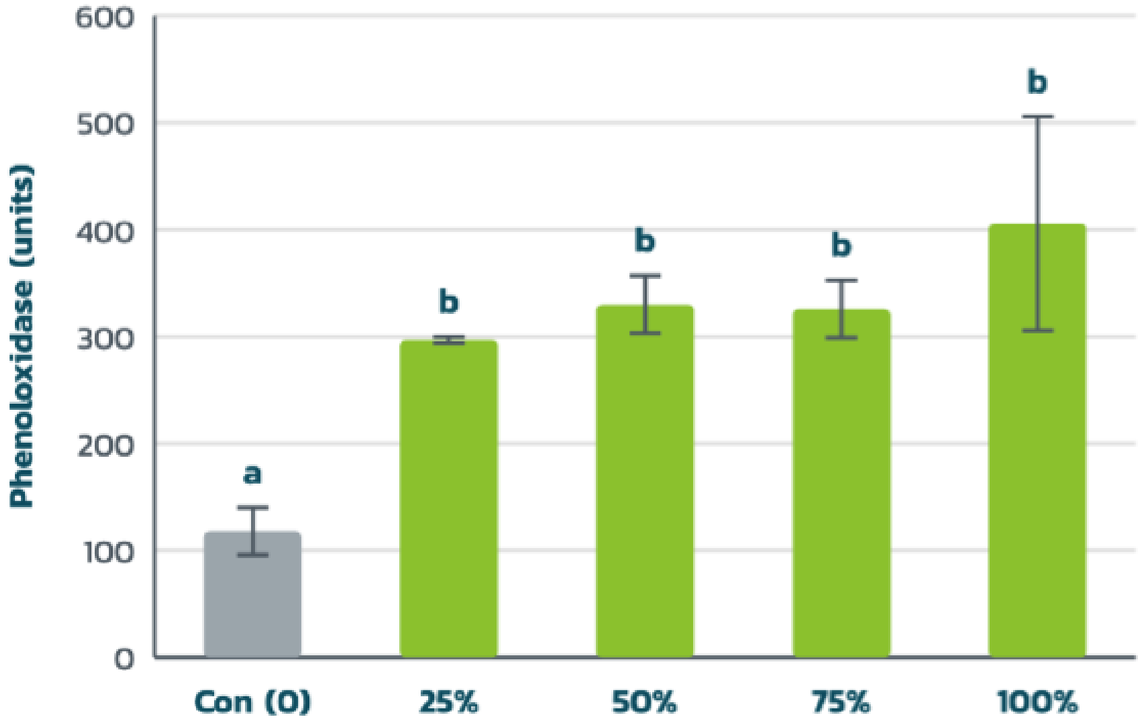
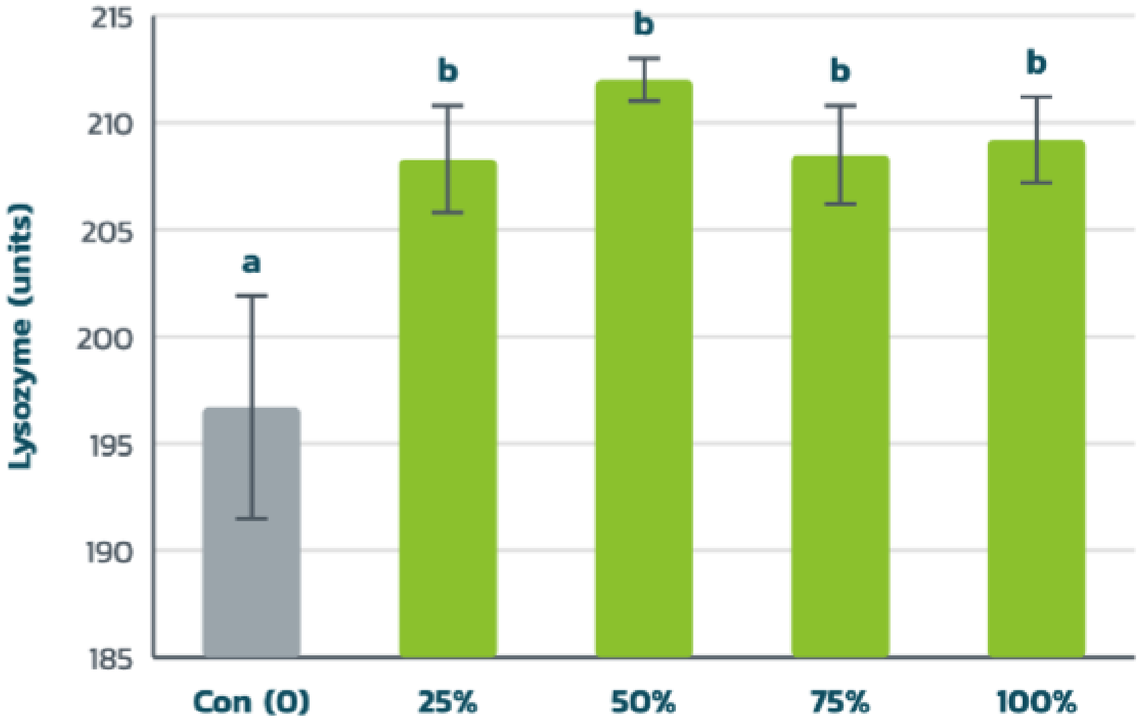


Figure 1. Histology of the intestinal villi of shrimp fed with pea starch concentrate as a replacement for wheat flour (WF). A – control group; B – shrimp fed 50% WF replacement; C – shrimp fed 100% WF replacement.

Shrimp fed the 100% wheat replacement diet developed villi approximately 39% taller and significantly narrower than those in the control group and the 50% replacement group.

IMMUNE RESPONSE





Prophenoloxidase and serum lysozyme were significantly increased in all pea starch concentrate (PSC) diets relative to the control group. Serum antibacterial activity was significantly enhanced at 75% and 100% wheat flour replacement, while total haemocyte counts were unaffected, indicating immune activation rather than physiological stress.

PEA STARCH CONCENTRATE……

1. Pea Starch Concentrate (PSC) completely replaced Wheat Flour in the diet of White Leg shrimp without compromising growth and performance.
2. PSC enhanced the shrimp immune response by increasing phenol oxidase and serum lysozyme activities.
3. PSC influenced intestinal villi morphology at 100% inclusion by increasing villi length and decreasing villi width relative to the control group (100% Wheat Flour), potentially improving nutrient absorption and utilization.



CANADIAN PEAS.....

A sustainable crop.....an ingredient for the future.

- Canadian peas are a sustainable and readily available ingredient for use in Philippine animal and aqua feed programs.
- A good source of protein and energy that can be included at relatively high levels in swine and poultry rations.
- Local research results are available on the use of field peas and pea fractions.





FROM CANADIAN FIELDS TO PHILIPPINE FEEDS

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